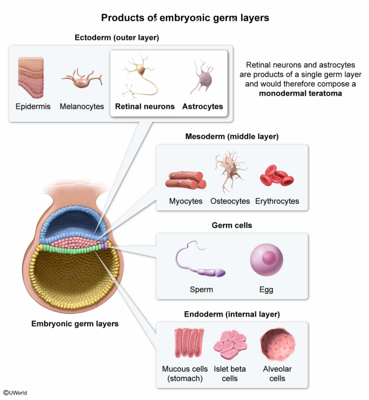
Teratomas are tumors with embryonic origin that differentiate to eventually form somatic tissues. Monodermal teratomas contain tissues derived from a single embryonic germ layer, whereas mature cystic teratomas contain tissues derived from at least two embryonic germ layers. If a teratoma is identified that contains astrocytes and retinal neurons, this teratoma is most likely a:

- A. monodermal teratoma, because astrocytes and retinal neurons both derive from mesoderm. [8%]
- ✔ B. monodermal teratoma, because astrocytes and retinal neurons both derive from ectoderm. [63%]
- C. mature cystic teratoma, because astrocytes derive from mesoderm and retinal neurons derive from ectoderm. [17%]
- D. mature cystic teratoma, because astrocytes derive from ectoderm and retinal neurons derive from mesoderm. [9%]

Correct
62% answered correctly

Explanation:



In most animal species, after an egg is fertilized and forms a zygote, it undergoes successive mitotic divisions (cleavage) and forms the blastula, a hollow ball of cells. The blastocyst, the human equivalent of the blastula, will ultimately implant in the wall of the uterus. The blastocyst is divided into two layers, the trophoblast (hollow cell sphere) and the inner cell mass (clustered inside the trophoblast).

The two-layer blastocyst then becomes a three-layered structure known as the gastrula during gastrulation, the process by which a developing embryo forms germ layers (ie, endoderm, mesoderm, ectoderm). Each of the three germ layers will continue to develop into specific body structures:

- **Endoderm** (innermost layer): organs that assist in digestion (eg, liver, pancreas) and the epithelium (lining) of the respiratory and digestive tracts
- **Mesoderm** (middle layer): musculoskeletal system, circulatory system, and portions of the reproductive and urinary systems
- **Ectoderm** (outermost layer): nervous system, integumentary system (skin, hair, nails), and lining of the nostrils, mouth, and anus

This question states that teratomas are tumors with embryonic origin that differentiate to eventually form somatic tissues. There are two types of teratomas in this example: monodermal teratomas, which contain tissues derived from a *single* embryonic germ layer, and mature cystic teratomas, which contain tissues derived from *at least two* embryonic germ layers.

If a teratoma is identified that contains astrocytes and retinal neurons (ie, nervous system cells), this teratoma contains structures found only in the nervous system. Because the nervous system is a product of a *single* germ layer (the ectoderm), astrocytes and retinal neurons both derive from the ectoderm. Therefore, this teratoma is most likely a **monodermal teratoma** because astrocytes and retinal neurons **both derive from the ectoderm** (*not* mesoderm), a single embryonic germ layer **(Choice A)**.

(Choices C and D) Because the structures found in this teratoma develop from only a single germ layer (ie, the ectoderm), this is an example of a monodermal teratoma, *not* a mature cystic teratoma, which contains tissue from at least two germ layers.

Educational objective:
The three primary germ layers (endoderm, mesoderm, ectoderm) form during gastrulation. Each gives rise to particular cell types in specific organ systems.

Time spent:0 sec
Subject:
Biology

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System:
Reproduction